

數統導論作業五

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4.1.2

(a)

The histogram of the 26 professional baseball pitchers' weights is shown in Figure 1.

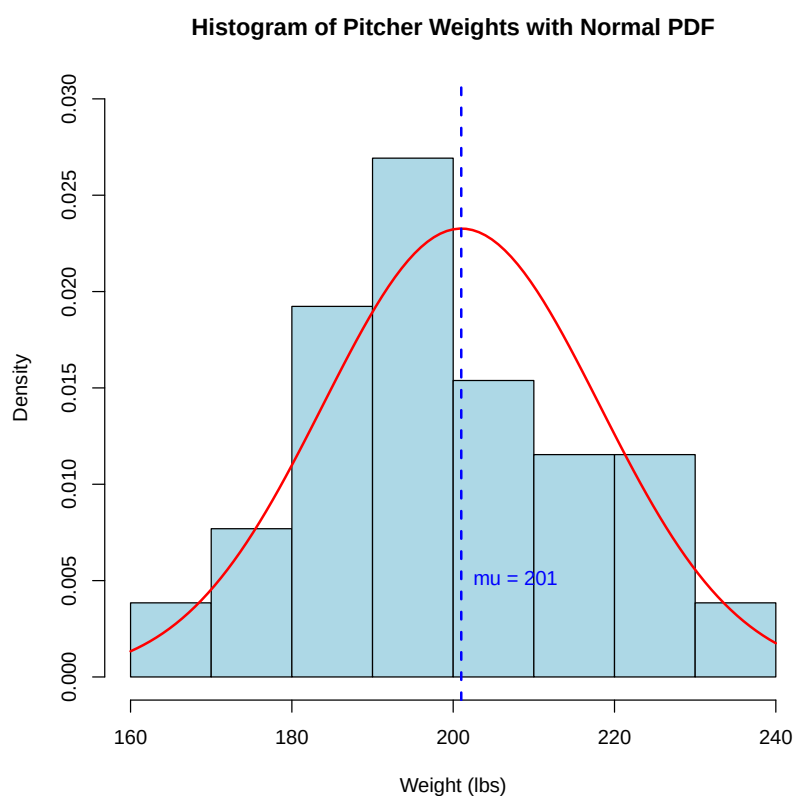


Figure 1: Histogram of pitcher weights with overlaid normal PDF.

Based on the plot, the data appears roughly symmetric and bell-shaped. Although the sample size ($n = 26$) is small, the normal probability model seems credible.

(b)

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution $N(\mu, \sigma^2)$. The probability density function is:

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}$$

The likelihood function is given by:

$$\begin{aligned} L(\mu, \sigma^2) &= \prod_{i=1}^n f(x_i; \mu, \sigma^2) \\ &= \prod_{i=1}^n \left((2\pi\sigma^2)^{-1/2} \exp \left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\} \right) \\ &= (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\} \end{aligned}$$

The log-likelihood function is:

$$l(\mu, \sigma^2) = \ln L(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

To find the MLEs, we differentiate $l(\mu, \sigma^2)$ with respect to μ and σ^2 , and set the derivatives to zero.

1. For μ :

$$\begin{aligned} \frac{\partial l}{\partial \mu} &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0 \\ \sum_{i=1}^n x_i - n\mu &= 0 \quad \Rightarrow \quad \hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X} \end{aligned}$$

2. For σ^2 :

$$\begin{aligned} \frac{\partial l}{\partial \sigma^2} &= -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 = 0 \\ \frac{n}{2\sigma^2} &= \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 \\ n &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \hat{\mu})^2 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2 \end{aligned}$$

Based on the data ($n = 26$):

- $\hat{\mu} = \frac{5226}{26} = 201$
- $\hat{\sigma}^2 = \frac{7642}{26} \approx 293.923$
- By the invariance property of MLE: $\hat{\sigma} \approx \sqrt{293.923} \approx 17.144$
- $\widehat{\mu/\sigma} \approx 201/17.144 \approx 11.724$

The estimate of μ is located at the center of the symmetry in Figure 1.

(c)

Let Y be the random variable representing the number of pitchers weighing over 215 pounds. We model Y using a binomial distribution $Y \sim \text{Bin}(n, p)$, where $n = 26$ and p is the unknown probability of success. The probability mass function (PMF) is given by:

$$P(Y = y) = \binom{n}{y} p^y (1 - p)^{n-y}$$

The likelihood function for the parameter p given the observation y is:

$$L(p) = \binom{n}{y} p^y (1 - p)^{n-y}$$

The log-likelihood function is:

$$\begin{aligned} l(p) &= \ln L(p) = \ln \left[\binom{n}{y} p^y (1-p)^{n-y} \right] \\ &= \ln \binom{n}{y} + y \ln p + (n-y) \ln(1-p) \end{aligned}$$

To find the maximum likelihood estimate $\hat{p}$, we differentiate $l(p)$ with respect to p and set the derivative to zero:

$$\begin{aligned} \frac{dl(p)}{dp} &= 0 + \frac{y}{p} - \frac{n-y}{1-p} = 0 \\ \frac{y}{p} &= \frac{n-y}{1-p} \\ y(1-p) &= p(n-y) \\ y - yp &= np - yp \\ y &= np \\ \hat{p} &= \frac{y}{n} \end{aligned}$$

Based on the data, the weights over 215 lbs are: 218, 219, 220, 222, 225, 225, 232.

- Number of successes (y): 7
- Total sample size (n): 26

Substituting these values into the MLE estimator:

$$\hat{p} = \frac{7}{26} \approx 0.2692$$

Thus, the estimated proportion of professional baseball pitchers who weigh over 215 pounds is approximately 26.9%.

4.1.3

(a)

Let $X_1, X_2, \dots, X_n$ be a random sample from a Poisson distribution $\text{Poisson}(\theta)$. The probability mass function is:

$$f(x; \theta) = \frac{e^{-\theta} \theta^x}{x!}$$

The likelihood function is given by:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n f(x_i; \theta) \\ &= \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!} \\ &= \frac{e^{-n\theta} \theta^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \end{aligned}$$

The log-likelihood function is:

$$\begin{aligned} l(\theta) &= \ln L(\theta) = \ln \left[\frac{e^{-n\theta} \theta^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \right] \\ &= -n\theta + \left(\sum_{i=1}^n x_i \right) \ln \theta - \ln \left(\prod_{i=1}^n x_i! \right) \end{aligned}$$

To find the maximum likelihood estimate $\hat{\theta}$, we differentiate $l(\theta)$ with respect to θ and set the derivative to zero:

$$\begin{aligned}\frac{dl(\theta)}{d\theta} &= -n + \frac{\sum_{i=1}^n x_i}{\theta} - 0 = 0 \\ n &= \frac{\sum_{i=1}^n x_i}{\theta} \\ n\theta &= \sum_{i=1}^n x_i \\ \hat{\theta} &= \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}\end{aligned}$$

Next, we show that $\hat{\theta}_{MLE}$ is an unbiased estimator:

$$E[\hat{\theta}_{MLE}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i]$$

Since $X_i \sim \text{Poisson}(\theta)$, we know that $E[X_i] = \theta$.

$$E[\hat{\theta}_{MLE}] = \frac{1}{n} \sum_{i=1}^n \theta = \frac{n\theta}{n} = \theta$$

We have proved that $E[\hat{\theta}_{MLE}] = \theta$. Therefore, the MLE of the Poisson distribution is an unbiased estimator.

(b)

Given the data, we calculate the estimate:

$$\hat{\theta}_{MLE} = \frac{1}{10}(9 + 7 + 9 + 15 + 10 + 13 + 11 + 7 + 2 + 12) = \frac{95}{10} = 9.5$$

This means that, on average, 9.5 customers enter the store between 9:00 a.m. and 10:00 a.m.

5.1.1

(1) Show that $a_n \rightarrow a \implies a_n \xrightarrow{P} a$

Assume $a_n \rightarrow a$.

1. By definition of $a_n \rightarrow a$ we have $\forall \epsilon > 0, \exists N$ that $\forall n > N$ we have $|a_n - a| < \epsilon$

2. Therefore, $\forall n > N$:

$$P(|a_n - a| \geq \epsilon) = P(\emptyset) = 0$$

3. Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} P(|a_n - a| \geq \epsilon) = 0$$

4. Thus, $a_n \xrightarrow{P} a$.

(2) Show that $a_n \xrightarrow{P} a \implies a_n \rightarrow a$

Assume $a_n \xrightarrow{P} a$.

1. By definition we have:

$$\forall \epsilon > 0, \lim_{n \rightarrow \infty} P(|a_n - a| \geq \epsilon) = 0$$

2. Note that since a_n is a constant sequence, the variable $|a_n - a|$ is deterministic. Thus, the probability $P(|a_n - a| \geq \epsilon)$ can only take values in $\{0, 1\}$.

$$P(|a_n - a| \geq \epsilon) = \begin{cases} 1 & \text{if } |a_n - a| \geq \epsilon \\ 0 & \text{if } |a_n - a| < \epsilon \end{cases}$$

3. A sequence taking values in $\{0, 1\}$ that converges to 0 must clearly be equal to 0 for all sufficiently large n .
4. Therefore, $\exists N \in \mathbb{N}$ such that $\forall n \geq N$:

$$P(|a_n - a| \geq \epsilon) = 0$$

5. This implies that for all $n \geq N$, the condition $|a_n - a| \geq \epsilon$ is false, which means:

$$|a_n - a| < \epsilon$$

6. This satisfies the definition of $a_n \rightarrow a$.

(3) Conclusion

We proved that:

$$a_n \xrightarrow{P} a \iff a_n \rightarrow a, \text{ when } a \in \mathbb{R}$$

5.1.2

(a)

Since $Y_n \sim b(n, p)$, we have :

$$\mathbb{E}[Y_n] = np, \text{Var}(Y_n) = np(1-p) \implies \mathbb{E}\left[\frac{Y_n}{n}\right] = p, \text{Var}\left(\frac{Y_n}{n}\right) = \frac{p(1-p)}{n}$$

By Chebyshev's inequality $\forall \epsilon > 0$:

$$P\left(\left|\frac{Y_n}{n} - \mathbb{E}\left[\frac{Y_n}{n}\right]\right| \geq \epsilon\right) \leq \frac{\text{Var}\left(\frac{Y_n}{n}\right)}{\epsilon^2} \implies P\left(\left|\frac{Y_n}{n} - p\right| \geq \epsilon\right) \leq \frac{p(1-p)}{n\epsilon^2}$$

Take limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{Y_n}{n} - p\right| \geq \epsilon\right) \leq \lim_{n \rightarrow \infty} \frac{p(1-p)}{n\epsilon^2} = 0$$

Also, we know that probability is a value in closed interval $[0, 1]$.

By squeeze theorem :

$$0 \leq \lim_{n \rightarrow \infty} P\left(\left|\frac{Y_n}{n} - p\right| \geq \epsilon\right) \leq \lim_{n \rightarrow \infty} \frac{p(1-p)}{n\epsilon^2} = 0 \implies \lim_{n \rightarrow \infty} P\left(\left|\frac{Y_n}{n} - p\right| \geq \epsilon\right) = 0, \quad \forall \epsilon > 0$$

Thus, Y_n/n converges in probability to p .

(b)

$\forall \epsilon > 0$, we have :

$$\lim_{n \rightarrow \infty} P\left(\left|\left(1 - \frac{Y_n}{n}\right) - (1-p)\right| \geq \epsilon\right) = \lim_{n \rightarrow \infty} P\left(\left|p - \frac{Y_n}{n}\right| \geq \epsilon\right) = \lim_{n \rightarrow \infty} P\left(\left|\frac{Y_n}{n} - p\right| \geq \epsilon\right)$$

By previous problem we have :

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{Y_n}{n} - p\right| \geq \epsilon\right) = 0, \quad \forall \epsilon > 0$$

Therefore, $\forall \epsilon > 0$:

$$\lim_{n \rightarrow \infty} P\left(\left|\left(1 - \frac{Y_n}{n}\right) - (1-p)\right| \geq \epsilon\right) = 0$$

Which means that $1 - Y_n/n$ converges in probability to $1 - p$.

(c)

By Slutsky's Theorem :

$$\frac{Y_n}{n} \xrightarrow{P} p \text{ and } 1 - \frac{Y_n}{n} \xrightarrow{P} 1 - p \implies \left(\frac{Y_n}{n}\right) \left(1 - \frac{Y_n}{n}\right) \xrightarrow{P} p(1 - p)$$

5.1.3

By Chebyshev's inequality, $\forall \epsilon > 0$:

$$P(|W_n - E[W_n]| \geq \epsilon) \leq \frac{\text{Var}(W_n)}{\epsilon^2} \implies P(|W_n - \mu| \geq \epsilon) \leq \frac{b}{n^p \epsilon^2}$$

Take limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} P(|W_n - \mu| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{b}{n^p \epsilon^2} = 0$$

Also, we know that probability is a value in closed interval $[0, 1]$.

By squeeze theorem :

$$0 \leq \lim_{n \rightarrow \infty} P(|W_n - \mu| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{b}{n^p \epsilon^2} = 0 \implies \lim_{n \rightarrow \infty} P(|W_n - \mu| \geq \epsilon) = 0, \quad \forall \epsilon > 0$$

Thus, W_n converges in probability to μ .

5.1.4

Let $Y_n = \max\{X_1, \dots, X_n\}$. By the definition of the CDF:

$$\begin{aligned} F_{Y_n}(t) &= P(Y_n < t) \\ &= P(\max(X_1, X_2 \dots X_n) < t) \\ &= P(X_1 < t, X_2 < t \dots X_n < t) \end{aligned}$$

Since $X_1, \dots, X_n$ are independent and identically distributed with CDF $F_X(t)$, we have:

$$F_{Y_n}(t) = \prod_{i=1}^n P(X_i < t) = [F_X(t)]^n$$

Substituting the CDF of the Uniform(0, θ) distribution :

$$F_X(t) = \begin{cases} 1 & t > \theta \\ \frac{t}{\theta} & 0 < t \leq \theta \\ 0 & t \leq 0 \end{cases}$$

We obtain:

$$F_{Y_n}(t) = \begin{cases} 1^n = 1 & t > \theta \\ \left(\frac{t}{\theta}\right)^n & 0 < t \leq \theta \\ 0^n = 0 & t \leq 0 \end{cases}$$

Therefore, we provd that :

$$F_{Y_n}(t) = \begin{cases} 1 & t > \theta \\ \left(\frac{t}{\theta}\right)^n & 0 < t \leq \theta \\ 0 & t \leq 0 \end{cases}$$

5.2.1

Since $X_1, \dots, X_n$ is a random sample from $N(\mu, \sigma^2)$, the sample mean has the distribution $\bar{X}_n \sim N(\mu, \sigma^2/n)$. The moment generating function (MGF) of $\bar{X}_n$ is:

$$M_{\bar{X}_n}(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2n}\right)$$

Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} M_{\bar{X}_n}(t) = e^{\mu t}$$

Since $e^{\mu t}$ is the MGF of a degenerate distribution with all probability mass at μ , the limiting distribution of $\bar{X}_n$ is a degenerate distribution at μ .